

Product Design

Product Design Road Map

The **Product Design Road Map** establishes that product and service design must be **collaborative and holistic**, involving all functions and supply chain partners affected by the design. Design is not limited to engineering aesthetics or functionality—it spans from marketing through production and distribution.

Key ideas:

- Design converts **customer needs or innovations** into usable products, processes, or services.
- Design decisions strongly influence supply chain performance:
 - Design costs are only ~5–15% of product cost
 - About **70% of delivery cost** is locked in by design choices
- Product traits affect:
 - Sourcing options
 - Manufacturing feasibility
 - Logistics efficiency
 - Sustainability and returns

Good design balances **functionality, validity (value relative to price), efficiency, quality, serviceability, and recyclability**. Poor design can make products uncompetitive due to high cost or slow time to market.

Traditional Design Process

The traditional (or “over-the-wall”) design process is **sequential and siloed**.

How it typically works:

- Marketing defines requirements and passes them to engineering
- Engineering designs the product and passes it to production and purchasing
- Production and purchasing discover cost, sourcing, or manufacturability issues
- Designs are repeatedly sent back for revision
- Logistics is involved late and may find packaging or capacity problems

This process often results in:

- Increased costs
- Longer design cycles
- Problems unknowingly built into the product
- Poor alignment with supply chain capabilities

Collaborative Design Process

Collaborative design replaces silos with **cross-functional and cross-organizational teams**.

Key characteristics:

- Design teams include:
 - Engineering
 - Marketing
 - Production
 - Purchasing
 - Supply chain partners (suppliers and sometimes customers)
- Supply chain implications are considered early, while changes are inexpensive
- Partners help estimate cost, feasibility, and tradeoffs during design

The result is:

- Faster time to market
- Lower total cost
- Better alignment with customer needs
- Fewer late-stage redesigns

How Much Collaboration?

Collaboration exists on a continuum:

1. **Over-the-wall** — no external input
2. **Informal collaboration** — advice via conversations
3. **Formal collaboration** — suppliers/customers are part of the design team

Research shows:

- Greater supplier and customer involvement generally leads to:
 - Lower cost
 - Higher quality
- The owning firm should **retain final design authority**, even when collaborating

Implementing Design Collaboration

Collaboration does not happen automatically—it must be **intentionally implemented**.

A successful approach includes:

- Proof-of-concept projects to demonstrate benefits
- Formalizing terminology, training, and expectations
- Creating cross-functional teams with assigned time and metrics
- Aligning incentives so teams benefit from supply chain cost reductions

This phased approach helps organizations adopt collaboration sustainably.

Benefits of Design Collaboration

Key benefits include:

- Fewer cost overruns
- Better manufacturability and sourcing feasibility
- Access to supplier expertise and innovation
- Improved customer satisfaction
- Shorter development cycles
- Higher quality at an acceptable price

Broad-Based Design Methods

Broad-based design approaches look at **overall system performance**, not isolated functions. The three main methods are:

- Design for the supply chain
- Design for logistics
- Design for X (DFX)

Design for the Supply Chain

Design for the supply chain means designing products with **end-to-end supply chain impact in mind**, from raw materials through end-of-life.

Goals:

- Faster inventory turnover
- Lower inventory and transit costs
- Reduced lead times
- Improved time to market

It integrates considerations related to:

- Standardization
- Simplification
- Customization
- Quality
- Sustainability

Design for Logistics (DFL)

Design for logistics focuses on **minimizing transportation, storage, and handling costs** through product and packaging design.

Key principles:

- Efficient packaging and pallet utilization
- Unitization and containerization
- Standard box sizes
- Reduced package volume and weight

Examples:

- Stackable plastic bins
- Flat-pack furniture (e.g., IKEA)
- Nested products

Benefits:

- Lower logistics and warehousing costs
- Better storage capacity utilization
- Faster handling and replenishment

Tradeoffs:

- Larger pallet quantities may not suit all retailers
- Weight vs. cube constraints must be balanced

Design for X (DFX)

Design for X (or Design for Excellence) is a **holistic design philosophy**.

It means designing for whichever "X" factors are strategically important, such as:

- Manufacturability
- Assembly
- Quality
- Service
- Reverse logistics
- Sustainability
- Customization

Each organization defines which "X" priorities matter most.

Standardization and Modularization

Standardization reduces variation by using **common designs, parts, or processes**. Modularization builds on this by creating **standard modules** that can be assembled into different products.

Benefits:

- Lower design and manufacturing costs
- Economies of scale
- Faster product development

Modular Design

Modular design uses standardized components to create product families.

Examples:

- Computers (RAM, drives, GPUs)
- Automobiles
- Furniture kits

Benefits:

- Reduced manufacturing and design cost
- Easier customization
- Faster assembly and logistics

Tradeoffs:

- Higher per-unit cost than fully integral designs
- Risk of assembly errors
- Less emphasis on aesthetic integration

Component Commonality

Component commonality uses **one part instead of many similar parts**.

Benefits:

- Lower purchasing and inventory costs
- Simplified production
- Reduced error risk

Tradeoffs:

- Design flexibility may decrease
- Potential quality compromises

Universality

Universality designs products for **multiple markets from the start**.

Examples:

- Dual-voltage power supplies
- One-size-fits-all products

Benefits:

- Increased sales volume
- Lower design and production costs

Tradeoffs:

- Less optimized for specific markets
- Potentially shorter product life cycles

Simplification

Simplification removes unnecessary complexity from products and services.

Benefits:

- Shorter lead times

- Higher quality
- Lower cost
- Better profitability

Key simplification approaches:

- Concurrent engineering
- Design for manufacture and assembly (DFMA)
- Design for service

Concurrent Engineering

Concurrent engineering involves **multiple stakeholders early** in design.

Benefits:

- Shorter development cycles
- More realistic designs

Tradeoffs:

- Largely replaced by more robust methods like DFMA

Design for Manufacture and Assembly (DFMA)

DFMA ensures products are:

- Easy to manufacture
- Easy to assemble
- Affordable
- Fast to bring to market

Goals:

- Reduce number of parts
- Minimize handling
- Avoid tight tolerances
- Simplify testing and assembly

Benefits:

- Lower cost
- Less variation and confusion
- Enables lean and modular strategies

Tradeoffs:

- May conflict with marketing or aesthetic preferences

Design for Service (Maintainability)

Design for service focuses on **easy maintenance and repair**.

Benefits:

- Lower total cost of ownership
- Higher customer satisfaction
- Opportunities for parts and service revenue

Tradeoffs:

- May compete with aesthetics or initial cost goals

Quality, Customization, and Sustainability

Quality

Quality in product design is defined as conformance to requirements or fitness for use. It is a central design concern because it directly affects customer satisfaction, cost, and competitiveness.

Key ways quality is defined:

- **Transcendent** – an ideal state of excellence
- **Product-based** – based on measurable attributes
- **User-based** – fitness for customer use
- **Manufacturing-based** – conformance to specifications
- **Value-based** – level of excellence relative to price

Quality also has two major components:

- **Quality of conformance** – absence of defects
- **Quality of design** – how well the product's features satisfy customers

Quality must be deliberately designed into products, not inspected in later.

Design for Quality

Design for quality focuses on using **measures and targets** to ensure the product meets customer needs in performance, aesthetics, and cost.

Key points:

- Uses both objective and subjective measures
- Organizations define what is "acceptable" vs. "exceptional"
- Quality is often an order qualifier, and in some strategies, an order winner

Benefits:

- Fewer defects and less waste
- Higher customer satisfaction

Tradeoffs:

- Higher upfront design and development costs

- Long-term savings can be difficult to directly attribute

Design for Six Sigma

Design for Six Sigma (DFSS) designs products and processes that meet very high quality levels (six sigma performance) while remaining cost-effective.

Key characteristics:

- Strong focus on the **voice of the customer (VOC)**
- Prevents defects through design rather than correction
- Closely linked to continuous improvement efforts

Its primary objective is **predictable, low-variation performance**.

Quality Function Deployment (QFD)

QFD is a structured methodology that translates customer needs into technical design requirements.

Key features:

- Uses tools such as the **House of Quality**
- Compares customer needs against:
 - Design characteristics
 - Competitor offerings
- Uses group decision-making to resolve tradeoffs with the customer's priorities in mind

Benefits:

- Improves customer service
- Helps prioritize conflicting requirements
- Provides all benefits of design for quality

Tradeoffs:

- Complex to implement
- Requires strong organizational commitment and discipline

Customization

Customization enables products or services to adapt to different **customer needs over time**. Pure engineering-to-order customization is expensive and slow, so organizations use structured methods that maintain efficiency. The three main customization approaches are:

- Postponement
- Mass customization
- Globalization

Postponement (Delayed Differential)

Postponement delays final product differentiation until **demand is known**, shifting customization closer to the customer.

Key characteristics:

- Produces generic or modular products first
- Final configuration happens late in the process
- A classic **push-pull supply chain strategy**

Benefits:

- Reduces inventory risk and bullwhip effect
- Lowers safety stock and pipeline inventory
- Improves cash flow
- Allows localized sourcing for last-stage components

Tradeoffs:

- Requires redesign of products, processes, and packaging
- May increase cost if product variety is low

Mass Customization

Mass customization combines **economies of scale** with **individual product differentiation**.

Key characteristics:

- Uses standardized modules
- Final assembly or configuration occurs close to the customer
- Often supported by postponement

Examples:

- HP printers assembled regionally
- Customized education programs, investment portfolios, and meals

Benefits:

- Lower inventory costs through demand aggregation
- Increased sales due to customer choice
- Efficient use of labor and skills at the point of differentiation
- Creation of local semiskilled jobs

Tradeoffs:

- Investment in equipment and training
- Potential tension with distributors
- Risk of quality issues if assembly is poorly designed or trained

Glocalization

Glocalization blends global design with **local adaptation**.

Key characteristics:

- Products are designed for global markets
- Modified to meet local:
 - Regulations
 - Cultural preferences
 - Customer expectations

Examples include:

- Region-specific fast-food menus
- Medical devices designed for emerging markets
- Media and entertainment tailored to local audiences

Reverse Innovation

Reverse innovation recognizes that:

- Innovations for emerging markets can succeed in developed markets
- Lower-cost, simpler solutions can create new markets

Key insights:

- Emerging markets do not evolve the same way as developed ones
- Products designed for affordability and access can disrupt mature markets

This challenges traditional assumptions about global product design.

Sustainable Products and Services

Sustainability-focused design considers the **entire product life cycle**, including environmental, social, and regulatory impacts.

Three major design approaches are:

- Design for the environment
- Design for reverse logistics
- Design for remanufacture

Design for Environment (DFE)

DFE integrates health, safety, and environmental factors early in design.

Key considerations:

- Reuse and recycling
- Reduced energy consumption
- Use of lighter materials
- Avoidance of hazardous substances

Benefits:

- Improved corporate reputation
- Lower legal and environmental liability
- Alignment with sustainability regulations and values

Tradeoffs:

- Higher manufacturing costs
- Potential compromises in convenience, durability, or safety

Design for Reverse Logistics

Design for reverse logistics focuses on returns, repairs, recycling, and warranties.

Key design implications:

- Clear instructions to prevent user error
- Easy disassembly for repair
- Reusable or returnable packaging
- Accessible customer support

Benefits:

- Lower return costs
- Increased customer loyalty
- Better insight into product design flaws

Tradeoffs:

- Reverse flows are complex and costly
- Cannot always use the same infrastructure as forward logistics

Design for Remanufacture

Design for remanufacture enables components to be reclaimed and reused in new products.

Key points:

- Captures value from material reuse
- Reduces dependence on raw materials
- Supports closed-loop supply chains

Examples:

- Heavy equipment manufacturers reclaiming and reselling rebuilt components

Benefits:

- Lower cost to customers
- Reduced environmental impact
- Lower product development costs

Tradeoffs:

- Cash tied up in returned inventory
- Remanufactured goods cannot be sold as "new" in some markets